SAI KOWSHIK ANANTHULA

askowshik@outlook.com | +1 (470)-815-6620 | Austin, TX | https://linkedin.com/in/ask7 | https://saikowshik007.github.io

EXPERIENCE

Oracle February 2026 - Present

Senior Member of Technical Staff *Austin, TX*

- Designed and delivered 40+ distributed control-plane APIs for autonomous Oracle Cloud Infrastructure workflows, covering resource lifecycle management, orchestration, state retrieval, cancellation, and validation across multiple backend services.
- Spearheaded persistence, controller, accessor, and validation layers for complex backend APIs, connecting service contracts to asynchronous durable state and operational workflows.
- Implemented idempotent mutation workflows using retry tokens, ensuring duplicate client requests could be safely retried without creating inconsistent resource state.
- Applied optimistic-concurrency principles to resource updates, using version and action-clock semantics to prevent stale writes and eliminate unnecessary database validation logic by more than 50%.
- Built resilient service interfaces using gRPC/API contracts, clearly defining request-response behavior, failure handling, error codes, and integration patterns for downstream consumers.
- Designed infrastructure data models and evolved them into explicit parent-child relationships, improving orchestration semantics, queryability, and long-term maintainability.
- Authored comprehensive API specifications for a growing platform surface, creating consistent resource contracts, error behavior, and integration guidance that reduced ambiguity for developers and stakeholders.
- Created developer tooling that connects source control, documentation, and issue tracking context to AI-assisted workflows using model context protocols, accelerating investigation, code reviews, design decisions, and root-cause analysis.
- Developed more than 20+ codex AI skills ranging from autonomous debugging of production issues, peer reviews, correctness and consistency of generated code with repository that improved developers efficiency by 60% observed over 1 month.

International Business Machines (IBM) January 2024 - February 2026

Software Engineer (IBM Verify) *Austin, TX*

- Designed and built a distributed real-time notification pipeline using AWS IoT Core and Apache Kafka, reducing latency by 30% and reliably processing over 1M daily events.
- Developed scalable REST APIs and optimized PostgreSQL and MSSQL schemas, achieving 99.99% uptime and sub-100ms latency through advanced query tuning and connection pooling strategies.
- Drove OpenShift Container Platform migration across US and India teams, implementing CI/CD pipelines and automated testing to cut deployment time by 60% and downtime by 40%.
- Served on-call for IBM Verify, triaging and resolving 40+ production incidents weekly while meeting 99% SLA compliance, reducing MTTR by 10%, and improving recurring-incident resolution through log analysis, troubleshooting, and stakeholder coordination.
- Improved system resilience and throughput by introducing shared subscription models and load balancing, reducing message processing overhead by 50%.
- Contributed to the upgrade of IBM MaaS360 VPN infrastructure, achieving 20% performance gains through network-level optimizations with zero-downtime deployment.
- Delivered a mission-critical Digital Badge platform projected to save $16M annually by replacing legacy systems with scalable backend services, improving physical access workflows for 350K+ employees.
- Collaborated with cross-functional teams and external clients to define requirements and present technical solutions to C-level executives.

Global Payments Inc January 2023 - December 2023

Software Engineer *Columbus, GA*

- Orchestrated an Apache Airflow and Kubernetes pipeline for 5M+ daily transactions, increasing throughput by 60% via parallelized Docker containers and Kafka partitioning.
- Built enterprise Java and Spring Boot applications and RESTful services, improving data-processing speed by 30% and system efficiency by 50%.
- Engineered a low-latency client transaction pipeline using Airflow, Kubernetes, Kafka, and Docker, achieving a 60% increase in batch-processing efficiency.
- Used AWS EC2, S3, and Lambda to migrate mainframe processing to event driven architecture by improving resilience by 50% and reduce operational costs by 30%.
- Implemented Jenkins CI/CD pipelines that accelerated deployment cycles by 40%, while deploying AWS services with 20% lower operational costs and 30% faster releases.
- Pioneered dynamic, responsive React interfaces, reducing page-load times by 50% and increasing user engagement by 35%.

Global Payments March 2022 - August 2022

Software Engineer Intern *Atlanta, GA*

- Developed Java and Spring Boot microservices and RESTful APIs that improved data-processing speed by 30% and system efficiency by 50%.
- Developed POJOs and DAOs for 80+ database entities using Spring JDBC and MyBatis, optimizing backend connectivity in a Controller-Service layered architecture.
- Launched backend APIs and frontend UI with Java, React, Spring Boot, EC2, S3, and REST APIs, improving mainframe-pipeline scalability and robustness by 80%.

Georgia State University August 2021 - December 2022

Graduate Software Engineer (Research) *Atlanta, GA*

- Conducted experiments to improve scalable automobile safety with object detection and IoT sensors, reducing overall system power consumption by 30%.
- Developed a collision-detection algorithm that outperformed baseline approaches and collected test results using Python, JavaScript, SQL, and Java.

PROJECTS

Job Agent AI *(React, FastAPI, PostgreSQL, OpenAI GPT-4, Docker, AWS)*

- Built a responsive React frontend with real-time status tracking, PDF generation, and Simplify.jobs integration for an AI-powered job application platform.
- Architected a FastAPI backend with PostgreSQL, multi-layer caching, and asynchronous resume-generation processing.
- Implemented ATS keyword matching and job-specific content tailoring, and deployed the platform with Docker and comprehensive API documentation.

IBM -- Power Patrol *(Python, IBM Cloud, XGBoost, Streamlit, FastAPI)*

- Designed and deployed a real-time Streamlit ML dashboard for server-resource pattern analysis and predictive infrastructure planning.
- Built a modular Gaussian Mixture Model and XGBoost pipeline to segment server-usage behavior and predict resource needs.
- Engineered custom feature engineering across 18+ metrics, model persistence, and real-time anomaly detection using Pandas, scikit-learn, and Plotly.
- Delivered team- and organization-level insights through interactive visualizations, config-driven analysis, and seamless model retraining.

EDUCATION

Georgia State University, M.S. Computer Science (GPA: 3.9) August 2021 - May 2023

GITAM University, B.S. Computer Science (GPA: 3.9) June 2017 - May 2021

SKILLS

Languages: Java, Python, JavaScript, SQL

Backend: Spring Boot, FastAPI, REST APIs, gRPC, Microservices, Distributed Systems, Event-Driven Architecture, Asynchronous Processing, Idempotency, Optimistic Concurrency Control

Cloud, Infrastructure & DevOps: Oracle Cloud Infrastructure (OCI), AWS, EC2, S3, Lambda, Kubernetes, Docker, OpenShift, Jenkins, CI/CD, AWS IoT Core

Data & Messaging: Apache Kafka, Apache Airflow, PostgreSQL, Microsoft SQL Server, Spring JDBC, MyBatis, Data Modeling, Query Optimization, Connection Pooling

Testing & Operations: JUnit, Mockito, Load Balancing, Production Support, Incident Response, API Documentation

AI & Applied ML: OpenAI GPT-5, Prompt Engineering, LLM Application Development, AI Workflow Automation, XGBoost, Scikit-learn, Pandas, Anomaly Detection